Fiscal Year:	FY 2014	**Task Last Updated:**	FY 06/09/2014
PI Name:	Mueller, Holger Ph.D.		
Project Title:	QWEP Analysis and Optimization		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-0001	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
Start Date:	06/28/2012	**End Date:**	06/28/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kohel, James (Jet Propulsion Laboratory) Yu, Nan (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL 1458850		
Performance Goal No.:			
Performance Goal Text:			

The equivalence of gravity and accelerated motion is the basis of general relativity's description of gravity as a result of curved space and time. This description, however, is incompatible with the quantum theories of the electroweak and the strong interactions, in which space and time exist independently of the forces. It is therefore not surprising that unified theories of gravity and quantum mechanics are expected to contain tiny violations of the gravity-acceleration equivalence. Such violations would manifest themselves as variations between the acceleration of free fall for different types of matter. These variations are considered as being among the most promising areas to look for signals arising from a unification of general relativity and quantum mechanics. ESA's quantum weak equivalence principle (QWEP) mission seeks such signals with a strongly improved sensitivity of 10-15 or better. By using quantum interference of ultracold atoms, it will use quantum particles where previous tests used bulk matter.

Task Description:	The research proposed here will combine theory and experiment to maximize the sensitivity and scientific impact of QWEP, and minimize the risk. Using the standard model extension (SME) as a theoretical framework, we will study which species of atoms, what quantum states, and what type of analysis might lead to the largest and most interesting signals. The choice of species, however, is equally dependent on experimental considerations. Using our existing atom interferometers, we will experimentally compare different technologies for the matter-wave beam splitters that are suitable for the candidate species, and develop strategies to minimize systematic effects such as rotation and gravity gradients. Based on the experience of Jet Propulsion Laboratory (JPL) and the Space Sciences Laboratory (SSL) at UC Berkeley, we will match these technologies to the requirement of space operation. Thereby, this work will apply fundamental theory, atomic physics experiment, and principles of astronautics to a comprehensive study optimization and analysis of the QWEP mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We seek to understand two things: What is the difference between a quantum and classical test of the weak equivalence principle? What kind of quantum test would provide the most new information regarding Einstein's principle of equivalence? We have made progress on answering these questions have been pursued both theoretically and experimentally.
Task Progress:	We have continued and finished the analysis of nuclear corrections to enhance the sensitivity to equivalence principle (EP) violations. We find that these improve the sensitivity to all types of EP violations in the Standard Model Extension to levels of parts in 10 million and better, a level about ten times improved relative to prior knowledge. The results were published in "Equivalence Principle and Bound Kinetic Energy", Michael A. Hohensee, Holger Müller, and R. B. Wiringa, Phys. Rev. Lett. 111, 151102 (2013). We formed the Consortium for Ultracold Atoms in Space (CUAS) to further NASA's cold-atom research in space, in particular using the Cold Atom Lab (CAL). We also worked with NASA researchers (Nan Yu, Sheng-Wey Chiow, and Jason Williams of JPL) on a design study for a dedicated mission to test the EP with cold atoms on the ISS, called QTEST. We experimentally studied systematic effects in atom interferometers, as required for the planned missions. We developed atom interferometry as a test of fundamental connections between mass and time measurement, in particular as a mass standard in the planned revision of the international system of units [A clock directly linking time to a particle's mass. Shau-Yu Lan, Pei-Chen Kuan, Brian Estey, Damon English, Justin Brown, Michael Hohensee, and Holger Müller, Science, 339, 554 (2013)]
Bibliography Type:	Description: (Last Updated: 01/29/2019)
Articles in Peer-reviewed Journals	Hamilton P, Zhmoginov A, Robicheaux F, Fajans J, Wurtele JS, Müller (Mueller) H. "Antimatter interferometry for gravity measurements." Phys Rev Lett. 2014 Mar 28;112(12):121102. Epub 2014 Mar 25. PubMed PMID: 24724644 ; http://dx.doi.org/10.1103/PhysRevLett.112.121102 , Mar-2014
Articles in Peer-reviewed Journals	Hamilton P, Kim G, Joshi T, Mukherjee B, Tiarks D, Mueller H. "Sisyphus cooling of lithium." Physical Review A. 2014 Feb; 89(2):023409. http://dx.doi.org/10.1103/PhysRevA.89.023409 , Feb-2014
Articles in Peer-reviewed Journals	Harms J, Slagmolen BJ, Adhikari RX, Miller MC, Evans M, Chen Y, Mueller H, Ando M. "Low-frequency terrestrial gravitational-wave detectors." Physical Review D. 2013 Dec 15;88(12):122003. http://dx.doi.org/10.1103/PhysRevD.88.122003 , Dec-2013
Articles in Peer-reviewed Journals	Hohensee M, Mueller H, Wiringa RB. "Equivalence principle and bound kinetic energy." Phys Rev Lett. 2013 Oct;111(15):151102. http://dx.doi.org/10.1103/PhysRevLett.111.151102 , Oct-2013
Books/Book Chapters	Mueller M. "Quantum mechanics, matter waves, and moving clocks." in "Proceedings of the International School of Physics 'Enrico Fermi' 2013. Course 188--Atom Interferometry." Ed. G. Tino, M. Kasevich. Italian Physical Society, January 2014. , Jan-2014